VAISHNAVI PANCHAVATI

Mountain View, CA

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Software engineer focused on distributed systems and core infrastructure, with experience in C++, Python, and Go.

Education

University of Massachusetts Amherst **Feb 2023 – Jan 2025**

Master of Science in Computer Science, GPA: ***4.0/4.0*** *Amherst, MA*

National Institute of Technology Karnataka (NITK) **Aug 2014 – May 2018**

Bachelor of Technology in Electrical and Electronics Engineering, CGPA: ***9.26/10*** *Surathkal, Karnataka*

Technical Skills

Programming Languages: C++, Python, Go
Frameworks & Libraries: gRPC, Protocol Buffers, Boost.Asio, Pytest
Tools & Platforms: Docker, Kubernetes, AWS, Prometheus, Grafana, Git, Linux
Systems & Infrastructure: Distributed systems, consensus protocols, fault tolerance, scheduling, concurrency, control planes, observability

Open Source

Ray Project **Sept 2025 – Present**

Open Source Contributor, Ray Serve PRs *Remote*

- Authored ***15+ merged PRs*** in Ray Serve, improving control-plane reliability across scheduling, routing, autoscaling, and rollout safety.
- Designed and implemented ***gang scheduling and placement strategies*** in Ray Serve, enabling atomic multi-replica allocation for LLM serving workloads.
- Improved Ray Serve autoscaling with stateful custom policy support, consistent scaling semantics, and finer control over scale-to-zero behavior.
- Added observability for Serve routing and LLM serving paths through resolution latency metrics, capacity queue telemetry, and rate-limited fatal error logging.
- Diagnosed and fixed control plane correctness issues across crash recovery, autoscaling state isolation, and async request-context propagation.

Projects

Distributed Consensus Module | *Go* | Blog **Oct 2025 – Jan 2026**

- Implemented a Raft-based consensus module in Go with leader election, durable persistence, replicated logs, commit tracking, and snapshotting for fault-tolerant state replication.
- Built a strongly consistent KV store on top of Raft with safe client retries during leader changes, achieving ***∼1K ops/sec throughput and ∼100ms p99 latency*** with fault injection.

Model Serving Platform | *C++, Python, gRPC* | Blog **April 2025 – Aug 2025**

- Architected a distributed LLM serving platform with separate control and data planes, supporting streaming inference for ***1,000+ concurrent users at 100+ requests/sec.***
- Redesigned a latency sensitive distributed data path in C++ using asynchronous gRPC streaming and nonblocking I/O, eliminating Python runtime contention and ***improving p95 latency by ∼2.5×.***
- Deployed containerized vLLM replicas across AWS A10G GPU nodes and built a Prometheus observability stack for monitoring throughput, latency, and replica health.

Work Experience

Texas Instruments **Aug 2019 – May 2022**

Software Engineer *Bangalore, Karnataka*

- Implemented C++ features for the calibration subsystem of a 5G transceiver module, and developed functional test infrastructure that increased coverage by ***35%*** and caught integration failures earlier.
- Built a Python regression orchestration framework executing ***1,000+*** nightly tests, reducing manual effort by ***40%*** and shortening release validation cycles from days to hours.

Qualcomm **Sept 2018 – April 2019**

Software Engineer *Hyderabad, Telangana*

- Implemented C++ and Python components for LTE/5G physical layer protocols on resource-constrained devices, optimizing throughput and signal performance.
- Debugged performance corner cases in LTE/5G protocol software using log analysis, improving throughput and signal-to-noise ratio test outcomes by ***30%.***